

A Monte-Carlo Spin-Polarized Fusion Neutron Source for Stellarator Wall-Loading, and its Cross-Validation Against the Analytic Field-Period Method

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June 30, 2026

Abstract

Spin polarization of deuterium–tritium fuel both enhances the fusion rate and makes the 14.1 MeV neutron emission anisotropic about the local magnetic field, so the first-wall neutron load of a polarized reactor depends on the plasma shape, the field geometry, and the polarization mix together. A companion paper reduces this load to closed, differentiable form for weakly shaped quasi-axisymmetric (QA) stellarators by a field-period perturbation, valid while the geometric non-axisymmetry is small. Here we build the complementary tool: a Monte-Carlo spin-polarized fusion neutron *source* for OpenMC that samples the polarized birth direction exactly for an arbitrary polarization mix and arbitrary local field, carries no restriction on the plasma shaping, and, unlike the analytic method, ray-traces the true device geometry and admits neutron transport (scattering) on a realistic first wall. We give the emission kernel and its separation into a birth-direction distribution and a scalar rate factor, an exact stratified sampler for the two polarized angular shapes with a closed-form inverse-CDF cross-checked against rejection, and a robust local→global frame rotation. We verify the sampler in isolation—the sampled directionality reproduces the normalized kernels to 0.3% per bin with unit mean—and then cross-validate the full source against the analytic method in the free-streaming limit where the two must agree. On a gentle toy QA boundary ($\varepsilon_{\text{eff}} \approx 0.08$) the two methods agree to $\leq 1.8\%$ on all four walls of a square-cross-section torus; on a real, published QA equilibrium mined from the QUASR database ($\varepsilon_{\text{eff}} \approx 0.43$) they agree to $\leq 1.8\%$ on three walls, with a 4–6% residual isolated to the inner (inboard) wall. We show, by elimination and by a direct single-filament construction, that this residual is a limit of the analytic point-patch flux integral on a strongly curved wall under strong shaping—exact for an axisymmetric source and growing as the square of the shaping amplitude—and not a defect of the Monte-Carlo source, whose sampler, geometry, and transport are exact. The result delineates the division of labour between the two methods: the closed form for weakly shaped QA where gradients are cheap, and the present source for strongly shaped geometries and for the scattering problem that the free-streaming analytic cannot address.

1 Introduction

Spin-polarized fusion (SPF) aligns the reactant nuclear spins to raise the D–T reaction cross section by up to 50% and to redirect the angular distribution of the fusion products [4, 3, 1]. The neutron carries 80% of the D–T energy, so the redistribution of the 14.1 MeV neutrons directly reshapes

the first-wall neutron load—the quantity that sets the blanket, shield, and material lifetimes of a reactor—and the optimal polarization choice is a system decision that must be made against that load [1, 2]. Scoping studies quantify the load through the neutron wall load (NWL), the first-flight neutron current through a wall patch weighted by the neutron energy; it omits scattering but is inexpensive and is therefore the natural object for design iteration [2, 1].

The polarized neutron emission is not isotropic. For a chosen mixture of deuteron and triton spin projections the differential cross section is a fixed combination of two angular shapes referred to the local field direction $\hat{\mathbf{B}}$: a $\sin^2 \theta$ shape that peaks perpendicular to $\hat{\mathbf{B}}$ and a $\frac{1}{4} + \frac{3}{4} \cos^2 \theta$ shape that peaks parallel to it, where θ is the angle between the emitted neutron and $\hat{\mathbf{B}}$ [1]. Because $\hat{\mathbf{B}}$ follows the magnetic geometry, the wall load of a polarized plasma is a convolution of the emission kernel, the field, and the source distribution over the plasma volume against the wall geometry.

Two complementary routes evaluate this convolution. The first is analytic. For an axisymmetric device the toroidal source integral over a planar ring reduces to incomplete elliptic integrals, giving a fast, automatically differentiable NWL suited to shape and polarization optimization [1]. A stellarator source loop is non-planar, the period integral becomes hyperelliptic, and the closed form is lost; the companion paper restores it perturbatively for weakly shaped QA boundaries by expanding in the geometric non-axisymmetry, with a single-quadrature fallback where the expansion is slow. That method is powerful precisely where its controlling parameter—the geometric excursion of the boundary relative to the wall standoff, ε_{eff} —is small, and it is a free-streaming (uncollided) method by construction.

The second route, developed here, is Monte-Carlo. We build an SPF neutron source for OpenMC [6]: given a polarization mix, a birth position distribution over the plasma, and the local field, it samples the polarized birth direction and hands OpenMC a neutron that is then transported through the true device geometry. This route makes no assumption about the magnitude of the shaping, ray-traces the exact walls rather than approximating each patch as a point, and—its decisive advantage—admits full neutron transport, so the same source that reproduces the free-streaming NWL can be run with scattering on a CAD or constructive-solid-geometry (CSG) first wall to obtain the actual volumetric loads that a reactor blanket sees. The two methods are therefore not competitors but a division of labour: closed-form gradients where the boundary is weakly shaped, and the Monte-Carlo source where it is strongly shaped or where transport matters.

A Monte-Carlo source is only useful if it is demonstrably correct, and the natural check is that it reproduces the analytic NWL in the regime where the analytic method is itself exact—the free-streaming limit at small ε_{eff} . This paper is organized around that check and its one interesting failure. Section 2 states the emission kernel and separates it into a birth-direction distribution and a scalar rate factor. Section 3 gives the exact stratified direction sampler, the birth-position and field models, and the OpenMC realization. Section 4 verifies the sampler in isolation. Section 5 cross-validates the full source against the analytic method on a toy QA boundary and on a real published QA equilibrium. Section 6 isolates the one residual—on the strongly curved inboard wall of the strongly shaped case—and shows, by elimination and by direct construction, that it is a limit of the analytic point-patch flux integral and not of the Monte-Carlo source. Section 7 describes the transition to the transport problem, and Section 8 summarizes.

2 The spin-polarized emission source

We follow the parameterization of Ref. [1]. Let the deuteron spin projection fractions be d_+, d_0, d_- and the triton fractions t_+, t_- . The three collision-mode fractions are

$$a = d_+ t_+ + d_- t_-, \quad b = d_0, \quad c = d_+ t_- + d_- t_+, \quad a + b + c = 1, \quad (1)$$

with the unpolarized fuel at $a = b = c = \frac{1}{3}$. The differential cross section, referred to the local field direction $\hat{\mathbf{B}}$, is

$$\frac{d\sigma}{d\Omega} = \frac{\sigma_0}{2\pi} \left[\frac{3}{4} a \sin^2 \theta + \left(\frac{2}{3} b + \frac{1}{3} c \right) \left(\frac{1}{4} + \frac{3}{4} \cos^2 \theta \right) \right], \quad \cos \theta = \hat{\mathbf{B}} \cdot \hat{\Delta}, \quad (2)$$

where $\hat{\Delta}$ is the emitted neutron direction and σ_0 is the cross section in the $S = \frac{3}{2}$ combined-spin state. Integrating Eq. (2) over the sphere gives the total rate $\sigma_{\text{tot}} = \sigma_0 \eta$ with $\eta = a + \frac{2}{3}b + \frac{1}{3}c$; the unpolarized value is $\frac{2}{3}\sigma_0$, pure a (mode A) is σ_0 (a 50% enhancement), and pure c (mode C) is $\frac{1}{3}\sigma_0$. Modes B and C share the same angular shape $\frac{1}{4} + \frac{3}{4} \cos^2 \theta$ and differ only in rate: the emission of a polarized plasma is controlled by two angular functions, not three.

The essential structure for sampling is that the *birth direction* and the *total rate* separate. Collecting the two angular shapes,

$$I(\theta) = w_{\perp} \sin^2 \theta + w_{\parallel} \left(\frac{1}{4} + \frac{3}{4} \cos^2 \theta \right), \quad w_{\perp} = \frac{3}{4}a, \quad w_{\parallel} = \frac{2}{3}b + \frac{1}{3}c, \quad (3)$$

the normalized birth-direction density is $p(\theta, \phi) = I(\theta) / \int I d\Omega$ with ϕ uniform, and it depends only on $(w_{\perp}, w_{\parallel})$. The overall rate factor η multiplies the spatial source strength and does not enter the direction sampling. Keeping these two concerns separate is the central design choice: the sampler draws a correctly shaped direction for any (a, b, c) , and η is applied, if absolute loads are wanted, as a scalar weight on the birth-position distribution. For the free-streaming cross-checks of this paper the direction is all that matters, and the birth energy is taken monoenergetic at 14.1 MeV; a Ballabio-broadened Gaussian at a specified ion temperature is available for physical runs and is irrelevant to the angular verification.

3 Monte-Carlo realization

Exact stratified direction sampling. Because Eq. (3) is a nonnegative combination of two shapes, we sample the birth direction by stratification: with probability $W_{\perp} / (W_{\perp} + W_{\parallel})$ draw from the $\sin^2 \theta$ shape, otherwise from the $\frac{1}{4} + \frac{3}{4} \cos^2 \theta$ shape, where $W_{\perp} = w_{\perp} \int \sin^2 \theta d\Omega = \frac{8\pi}{3} w_{\perp}$ and $W_{\parallel} = w_{\parallel} \int (\frac{1}{4} + \frac{3}{4} \cos^2 \theta) d\Omega = 2\pi w_{\parallel}$ are the solid-angle-integrated weights. This is an exact draw from the mixture and makes per-mode verification trivial. Each shape is sampled by a closed-form inverse-CDF in $x = \cos \theta$: the $\sin^2 \theta$ shape has marginal $\propto \sin^3 \theta$ and is inverted by the trigonometric solution of the depressed cubic $x^3 - 3x + 2(1 - 2\xi) = 0$; the $\frac{1}{4} + \frac{3}{4} \cos^2 \theta$ shape has marginal $\propto \frac{1}{4} + \frac{3}{4} x^2$ and is a cubic in x as well. Each inverse-CDF is cross-checked against an independent rejection sampler (envelope $\propto \sin \theta$, acceptance $\frac{2}{3}$) to a two-sample Kolmogorov–Smirnov tolerance, so the root selection in the cubic is verified rather than assumed. The azimuth ϕ is uniform on $[0, 2\pi)$.

Local-to-global rotation. The sampled direction is expressed in a frame with $\hat{\mathbf{z}}_{\text{loc}} \parallel \hat{\mathbf{B}}$ and rotated to global Cartesian by a Gram–Schmidt construction that avoids the $\hat{\mathbf{B}} \parallel \hat{\mathbf{z}}$ degeneracy: the reference axis is whichever of $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ is least aligned with $\hat{\mathbf{B}}$, and $\hat{\mathbf{x}}_{\text{loc}} = \text{normalize}(\hat{\mathbf{e}}_{\text{ref}} - (\hat{\mathbf{e}}_{\text{ref}} \cdot \hat{\mathbf{B}}) \hat{\mathbf{B}})$, $\hat{\mathbf{y}}_{\text{loc}} = \hat{\mathbf{B}} \times \hat{\mathbf{x}}_{\text{loc}}$. The resulting global direction is unit to machine precision and, tested with a random non-axis $\hat{\mathbf{B}}$, leaves an isotropic local distribution isotropic—the check that the rotation carries no bias.

Birth positions and field. The birth-position model is pluggable. For the cross-checks the plasma is represented by constant- (ρ, ϑ) filaments on a set of scaled flux surfaces, sampled uniformly in the toroidal angle and weighted by the arc-length Jacobian and a model emissivity $W(\rho) \propto \rho^2$;

Table 1: Sampler self-test: sampled directionality A/iso and B/iso per $\cos\theta$ bin against the exact normalized kernels, and the mode means. The sampled kernels match to 0.3% and integrate to unit emission (1.0000), so the direction draw is exact independent of any geometry.

$\cos\theta$	A/iso (MC)	$\frac{3}{2}\sin^2\theta$	B/iso (MC)	$2(\frac{1}{4} + \frac{3}{4}\cos^2\theta)$
-0.95	0.146	0.146	1.857	1.854
-0.15	1.467	1.466	0.535	0.534
+0.05	1.491	1.496	0.505	0.504
+0.85	0.414	0.416	1.582	1.584
mean	1.0000	1	1.0000	1

the same filaments are the source loops of the analytic method, so the two calculations share the source exactly and the comparison isolates the transport treatment. The local field direction $\hat{\mathbf{B}}$ is evaluated at each birth point. For the real-equilibrium test we take the flux-surface-tangent field $\hat{\mathbf{B}} \propto \partial_\phi \mathbf{r} + \iota \partial_\theta \mathbf{r}$ at the device rotational transform ι , i.e. the field-line direction implied by the boundary geometry and the transform; because both the analytic and the Monte-Carlo evaluation call the identical $\hat{\mathbf{B}}$, the comparison is valid independent of how faithfully this model reproduces the true equilibrium field.

OpenMC source and transport. The sampled births are handed to OpenMC [6] either through a compiled custom-source shared library or, as used here, as a pre-sampled source bank written to an HDF5 file and read back as a fixed source. For the free-streaming cross-checks the vessel interior is void and the walls are vacuum boundaries, so each neutron streams in a straight line from its birth point until it crosses a surface and is tallied; the wall load is a surface-current tally binned poloidally. This ray-traced transport makes no point-patch or $1/D^2$ approximation—a neutron reaches a wall patch if and only if its sampled trajectory does—which is the property that the analytic comparison below tests. The same source, run with a material first wall and blanket rather than a void, gives the transported load; that calculation is the subject of Sec. 7.

4 Verification of the sampler

The sampler is verified before any geometry is introduced. Drawing 2×10^6 directions per mode about a fixed $\hat{\mathbf{B}} = \hat{\mathbf{z}}$ and histogramming in equal $\cos\theta$ bins, the sampled directionality—the ratio of the mode density to the isotropic density in each bin—reproduces the normalized kernels $\frac{3}{2}\sin^2\theta$ (mode A) and $2(\frac{1}{4} + \frac{3}{4}\cos^2\theta)$ (mode B/C) to better than 0.3% per bin, and the mean directionality of each polarized mode is 1.0000, confirming that the stratified draw is normalized to unit emission exactly (Table 1). A χ^2 test against the analytic $p(\cos\theta)$ for {unpolarized, A, B, C, mixed} and a two-sample KS test of inverse-CDF against rejection close the sampler verification; the isotropic and rotation-isotropy checks are flat as required. This isolates the sampler from every subsequent test: any later discrepancy is geometric, not a defect of the angular draw.

5 Cross-validation against the analytic method

The decisive test is that the Monte-Carlo source reproduces the analytic free-streaming NWL where the analytic method is itself exact. We compare on a shared square-cross-section torus wall, so that

Table 2: Toy QA ($\varepsilon_{\text{eff}} \approx 0.08$): directionality A/iso , B/iso as a ratio of total wall loads, Monte-Carlo vs analytic. All walls agree to $\leq 1.8\%$.

wall	A MC	A an	B MC	B an
inboard	1.207	1.229	0.782	0.771
outboard	0.823	0.829	1.182	1.171
floor	1.065	1.069	0.937	0.931
ceiling	1.062	1.066	0.937	0.934

Table 3: Real QUASR QA ($\varepsilon_{\text{eff}} \approx 0.43$): same comparison. Outboard/floor/ceiling agree to $\leq 1.8\%$; the inboard wall carries a 4–6% residual.

wall	A MC	A an	B MC	B an
outboard	0.833	0.848	1.165	1.152
floor	1.048	1.057	0.955	0.943
ceiling	1.043	1.057	0.954	0.943
inboard	1.248	1.199	0.755	0.801

all non-axisymmetry lives in the source and the wall is common to both calculations, and we compare the wall directionality A/iso and B/iso —the ratio of the polarized load to the isotropic load on each wall—which cancels the source normalization and isolates the polarized steering. The analytic side is a pure-numpy free-streaming quadrature that evaluates the loop geometry and field directly at the Gauss–Legendre nodes with the true shaped-loop visibility and the inner-cylinder occlusion; it is gated at run time against the companion **anarríma** quadrature and matches it to 10^{-12} , so it is the analytic method’s own reference, only faster.

A gentle toy QA boundary. The first case is a deliberately gentle QA boundary with $N_{\text{fp}} = 2$, unit major radius, minor radius ≈ 0.17 , and small non-axisymmetric harmonics giving a gently rotating-elliptical cross section, at $\varepsilon_{\text{eff}} \approx 0.08$. The source is filaments on two scaled flux surfaces with a ρ^2 emissivity and a model field carrying a mean pitch and a field-period ripple; the wall is the square torus at a fixed gap. With 10^6 births per mode the Monte-Carlo directionality matches the analytic on all four walls to $\leq 1.8\%$ (Table 2), and the per-bin poloidal profiles track—the correlation between the Monte-Carlo and analytic A/iso profiles is 0.85–0.98 on the walls that carry a poloidal gradient (Fig. 1). The gentle case therefore validates the whole chain—sampler, birth model, field, ray-traced transport, and tally—against the analytic method with no residual beyond Monte-Carlo statistics.

A real, published QA equilibrium. The second case uses a genuine published equilibrium. Because the standard published QA configurations are all strongly shaped—a survey of DESC examples gives $\varepsilon_{\text{eff}} \approx 1\text{--}2.6$ at a tight wall—we mined the QUASR stellarator database [8] for a gentle one without the bulk download, by pulling its single tabulated catalogue (3.7×10^5 devices), filtering to quasi-axisymmetry (7.2×10^4 devices), ranking by a metadata proxy that correlates with the geometric excursion (correlation 0.94), and computing the exact ε_{eff} of a shortlist from their VMEC boundary spectra. The gentlest is a $N_{\text{fp}} = 3$ QA of aspect ratio 6.7 (unit major radius, minor radius ≈ 0.175), which at the validation wall has $\varepsilon_{\text{eff}} \approx 0.43$ —four times more shaped than the toy and, unlike the toy, near the edge of the analytic method’s convergent band. Its rotating cross section is shown in Fig. 2.

On this configuration the Monte-Carlo and analytic directionalities agree to $\leq 1.8\%$ on the outboard, floor, and ceiling walls, with the poloidal profiles tracked to correlation 0.7–0.99; the inner (inboard) wall shows a larger, 4–6% residual (Table 3, Fig. 3). The inboard wall is where the analytic companion paper’s own controlling parameter ε_{eff} is largest—it is local and peaks where the standoff is smallest—so a departure there, on the most strongly curved wall of the most strongly shaped case, is exactly where one expects the free-streaming analytic to be most stressed. We now show that this residual is a limit of the analytic method, not of the Monte-Carlo source.

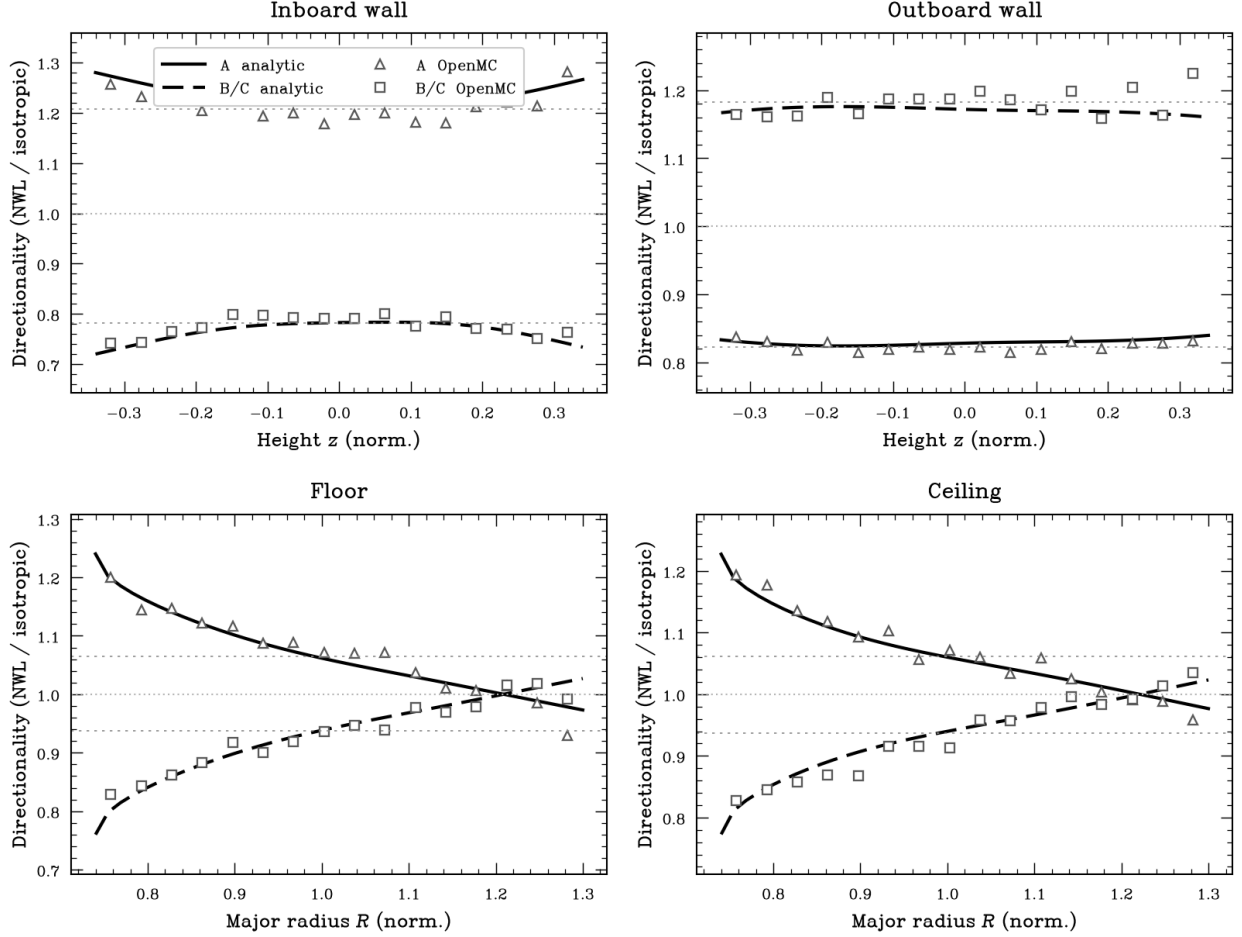


Figure 1: Free-streaming directionality on the gentle toy QA boundary ($\varepsilon_{\text{eff}} \approx 0.08$): analytic point-patch flux integral (curves) and OpenMC ray-traced source (markers) for modes A and B/C around each wall of the square torus. All four walls overlay to $\leq 1.8\%$; the poloidal profiles track to correlation 0.85–0.98 on the walls that carry a gradient.

6 The point-patch limit of the analytic method

The inboard residual is small but coherent, and its cause is worth establishing because it maps the boundary between the two methods. We first eliminate the Monte-Carlo source as the cause, then reproduce the effect by direct construction.

By elimination. The residual is not the direction sampler, which reproduces the kernels exactly (Sec. 4). It is not the analytic quadrature, which matches *anarríma* to 10^{-12} . It is not the visibility model: using the true shaped-loop $N > 0$ sub-arcs intersected with the inner-cylinder occlusion—rather than a mean-radius, uncapped arc—removes an unrelated outboard error but leaves the inboard value unchanged. It is not the field model, because the analytic and Monte-Carlo evaluations call the identical $\hat{\mathbf{B}}$, which cannot by itself open a gap between them. And it is not a near-field effect of the wall proximity: moving the wall outward from a gap of $0.4a$ to $0.7a$ leaves the offset unchanged ($0.036 \rightarrow 0.034$). What does change it is the plasma shaping—the gentle toy agrees on the inboard wall to 1.8%, the strongly shaped QA does not—so the effect is a

QUASR QA 59509 source loops (LCFS + $\rho = 0.6$) in the square wall

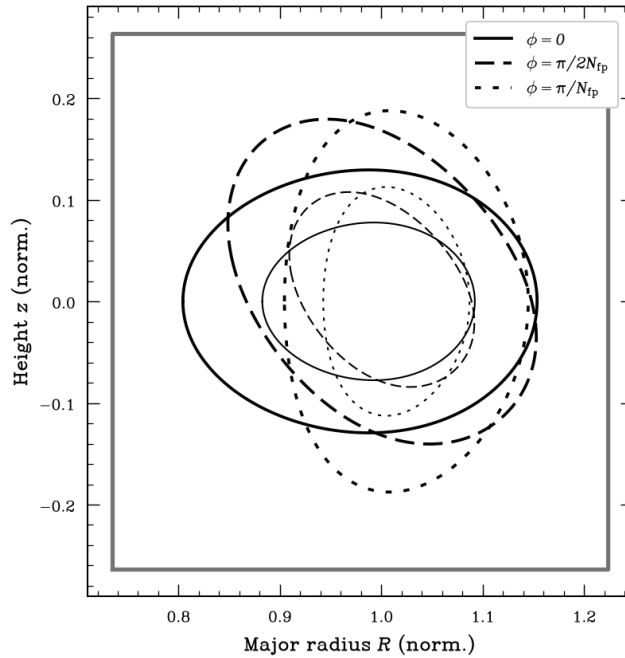


Figure 2: Source loops of the real QUASR QA equilibrium (device 59509, $N_{fp} = 3$): the last-closed and an interior flux surface at several toroidal cuts, showing the rotating, strongly shaped cross section, inside the square-cross-section wall used for the cross-check.

coupling of the source shaping to the wall geometry that the two methods treat differently.

By construction. The one geometric approximation in the analytic NWL is that each wall patch is a point: the emission kernel is evaluated along the single line of sight from the source to the patch centre, whereas a patch on a curved wall subtends a finite solid angle over which the kernel varies, and the ray-traced Monte-Carlo resolves it. To exhibit this in isolation we strip the problem to a single shaped filament with a tunable shaping amplitude s , emitting the kernel about a purely toroidal field onto bare inner and outer cylinders, and evaluate the wall directionality two ways: the analytic point-patch flux integral, and an exact ray/surface intersection of the same sampled births (the transport, with no point-patch approximation). Here the sampler, field, and occlusion are all trivially correct, so the shaping amplitude is the only variable.

The result (Fig. 4) is decisive. At $s = 0$, an axisymmetric source, the point-patch integral is *exact*: the analytic and ray-traced directionalities agree to 0.002, the Monte-Carlo statistical floor. As s increases the two diverge, with the discrepancy growing as s^2 , and it grows about three times faster on the convex inner cylinder (the inboard analogue) than on the concave outer cylinder (the outboard analogue), reaching $\sim 10\%$ on the inner wall at the strongest shaping. This reproduces, in a controlled single-filament setting, both the inboard-specific location and the shaping scaling of the residual seen on the full QA equilibrium, and it identifies the cause unambiguously as the point-patch treatment of the flux integral on a strongly curved wall—a property of the analytic method, with the geometry-exact, sampler-exact Monte-Carlo source as the reference. It is precisely the kind of strongly shaped, strongly curved regime that motivates a transport calculation in the first place.

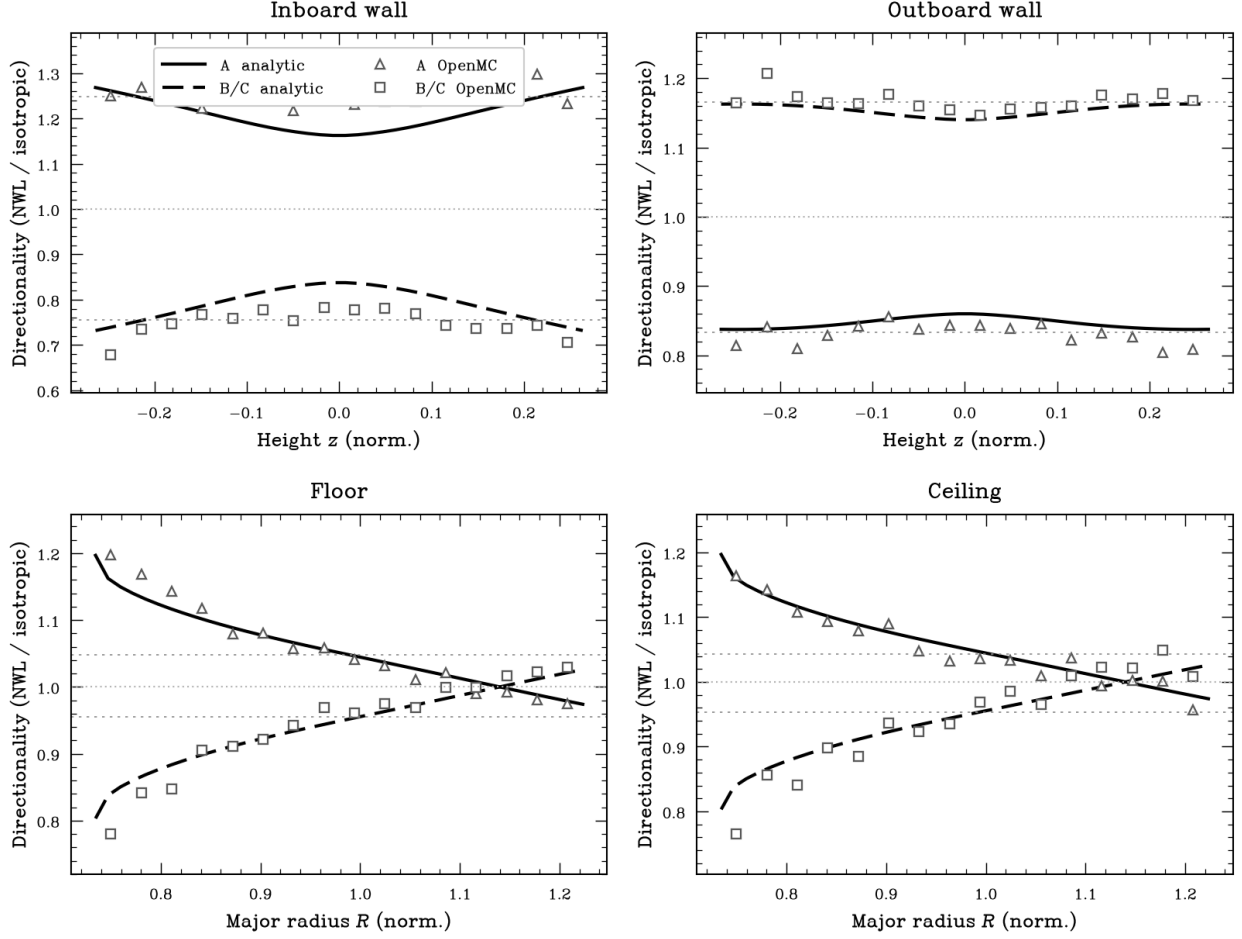


Figure 3: Free-streaming directionality A/iso (mode A) and B/iso (mode B/C) around each wall of the square torus for the real QUASR QA equilibrium ($\varepsilon_{\text{eff}} \approx 0.43$): the analytic point-patch flux integral (curves) and the OpenMC ray-traced source (markers). The outboard, floor, and ceiling walls overlay; the inboard wall tracks the poloidal shape with the 4–6% offset analyzed in Sec. 6. Per-panel vertical scales because the walls differ by an order of magnitude in poloidal curvature.

7 Toward transport on a realistic wall

The free-streaming cross-check certifies the source; the reason to have a Monte-Carlo source is what the analytic method cannot do. Two extensions follow directly and change no part of the sampler. The first is geometry: the same pre-sampled or compiled source drives a run on a CAD first wall imported through DAGMC [7] or on a CSG blanket model, so the polarized load is evaluated on the true toroidal-field-coil-limited wall shape rather than an idealized torus, and on the strongly shaped boundaries where the analytic expansion is out of its band. The second is physics: replacing the void interior and vacuum walls with a first wall, breeder, and shield turns the first-flight NWL into the transported neutron flux and heating that set the tritium breeding ratio and the blanket lifetime. The polarized angular emission enters that calculation only through the source; everything downstream is standard neutron transport, so the anisotropy is carried into the volumetric loads without approximation. These runs are most naturally performed where DAGMC and OpenMC install together against a common cross-section library; we report them separately.

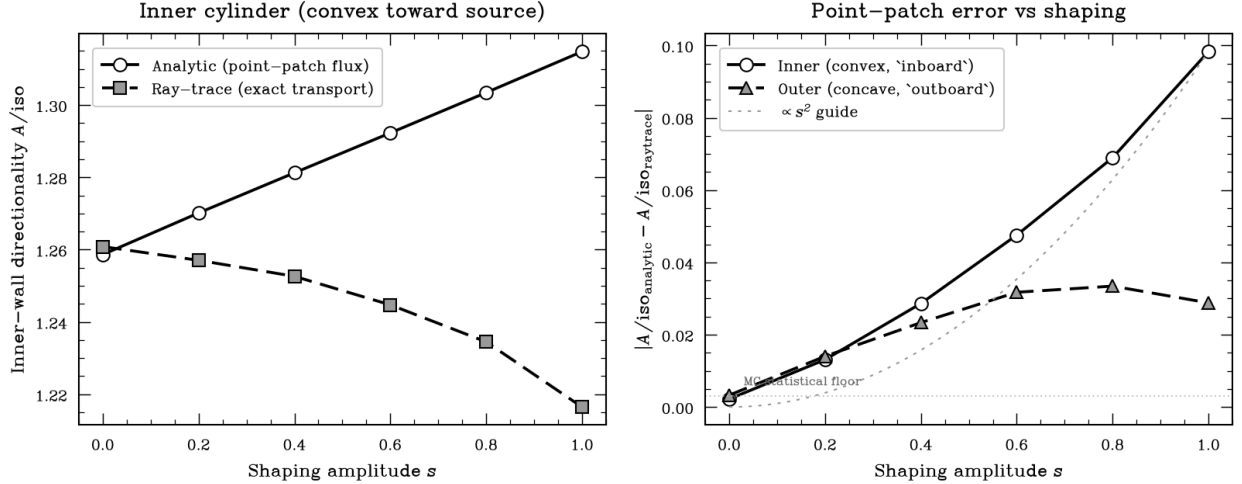


Figure 4: Single-filament demonstration that the inboard residual is the analytic point-patch limit. Left: inner-wall directionality A/iso from the analytic point-patch flux integral and from exact ray-tracing of the same births, versus the shaping amplitude s ; coincident for an axisymmetric source ($s = 0$) and diverging as s grows. Right: the analytic $\leftrightarrow$ ray-trace discrepancy versus s for the convex inner (inboard) and concave outer (outboard) cylinders, with an s^2 guide; zero to Monte-Carlo statistics without shaping, growing quadratically with shaping and largest on the convex inner wall.

Two limitations bound the present results. The birth model is filamentary or flux-surface-weighted rather than a self-consistent fusion reactivity profile, and the field is a model flux-surface-tangent field rather than a VMEC solution; both are shared exactly between the two methods, so they do not affect the cross-validation, but a physical run should replace them with a reactivity-weighted volumetric source and the equilibrium field. Neither touches the sampler, which is the object this paper verifies.

8 Summary

We have built and verified a Monte-Carlo spin-polarized fusion neutron source for OpenMC. The emission kernel separates cleanly into a birth-direction distribution, sampled exactly by stratification over the two polarized angular shapes with inverse-CDF draws cross-checked against rejection, and a scalar rate factor applied to the source strength; the local-to-global rotation is degeneracy-free and bias-free. The sampler reproduces the normalized polarized kernels to 0.3% with unit mean, independent of any geometry. Run as a free-streaming source on a shared wall, it reproduces the companion analytic method's NWL to $\leq 1.8\%$ on a gentle QA boundary on all walls, and to $\leq 1.8\%$ on three walls of a real, strongly shaped published QA equilibrium mined from the QUASR database. The one larger residual, 4–6% on the strongly curved inboard wall of the strongly shaped case, we have shown—by systematic elimination and by a single-filament construction that is exact for an axisymmetric source and grows as the square of the shaping amplitude—to be a limit of the analytic point-patch flux integral rather than a defect of the Monte-Carlo source. The two methods thus partition the problem cleanly: the closed, differentiable analytic form where the boundary is weakly shaped and gradients drive optimization, and the present source where the shaping is strong or where the neutron transport on a realistic wall—the calculation the free-streaming methods can-

not perform—is what a reactor design actually needs.

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